Nightly Maintenance Workflow

Each `0x12` command shall send the full command data portion. For readability, the workflow may only highlight the command fields relevant to the current step. Command fields not shown in a detailed command block shall follow the active command state required for that workflow step.

For `0x12` request changes, the Jetson shall resend the same command until either:

1. the LC acknowledges the command, or
2. the Jetson observes the expected state transition through `0x11` status polling.

Once the request change is acknowledged or the expected state transition is observed, the Jetson shall stop retrying that request change.

The `0x12` reply only confirms that the command frame was received and accepted by the LC. The Jetson shall monitor the actual progress and result of the automatic service session using `0x11`.

While robot maintenance operation remains active, the Jetson shall continue sending periodic `0x12` hold frames with:

Automatic Service Request = 1 (ON)
Automatic Service Request Type = 0x01

Repeated `0x12` frames with the same held request state shall be treated as the same active robot maintenance request, not as repeated new start requests.

During LC-controlled automatic service preparation, the Jetson shall not send Manual Up, Manual Down, destination floor, or door commands. However, while robot maintenance operation is active, the Jetson shall continue holding Automatic Service Request = ON through periodic 0x12 hold frames and shall continue polling LC status using 0x11.

During LC-controlled automatic service completion, after the Jetson has cleared Automatic Service Request to OFF, the Jetson shall not send Manual Up, Manual Down, destination floor, or door commands. The Jetson shall only poll LC status using 0x11 and wait for Service Complete / Normal Operation Restored.

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1. Scheduled Start

At `03:00` daily, the Jetson begins the automatic maintenance sequence.

The Jetson sends a `0x12` command to request the LC to start automatic service preparation.

```
Function Code: 0x12
Destination Floor Byte = 0x00
Automatic Service Request = 1 (ON)
Automatic Service Request Type = 0x01
Request Type Meaning = Start / Hold Robot Maintenance Operation
```

The Jetson shall continue holding Automatic Service Request = ON while the robot maintenance operation remains active.

After sending this command, the Jetson continues sending periodic `0x12` hold frames with `Automatic Service Request = ON` and polls the Lift Controller using `0x11`.

```
Function Code: 0x11
Purpose: Retrieve operational status bits and automatic service session
status
```

The LC shall handle the automatic service preparation sequence internally.

```
LC automatic service preparation responsibilities:
- Check lift readiness
- Disable hall calls / normal passenger operation
- Enter Robot Maintenance Mode
- Move/home the lift to the top service floor
- Report ready for Jetson/robot control
```

The Jetson shall not start robot operation during this stage. The Jetson continues sending periodic 0x12 hold frames with Automatic Service Request = ON and polls 0x11 while waiting for the LC to report that preparation is complete.

Expected LC status before Jetson-controlled robot operation can begin:

```
Automatic Service Session State = Ready for Jetson/Robot Control
Jetson Operation Permitted = 1 (ON)
Robot Maintenance Mode = 1 (ON)
Lift Abnormal Mode = 0 (OFF)
```

`Ready for Jetson/Robot Control` means:

```
- LC has completed automatic service preparation
- Lift is in Robot Maintenance Mode
- Hall calls are disabled
- Lift is stationary when control is handed over to the Jetson
- Doors are closed / no door movement is active
- Jetson Operation Permitted is ON
- Robot Maintenance Mode is ON
```

```
- Lift Abnormal Mode is OFF
- Jetson is allowed to perform manual down positioning and robot inspection
```

The lift is expected to be at the top service floor when `Ready for Jetson/Robot Control` is first reported after automatic service preparation. During Jetson-controlled inspection, the lift may move downward while this state remains active, as long as the LC still permits Jetson/robot control.

If the LC reports that the lift is not available for automatic service:

```
Automatic Service Session State = Not Available
```

Then the Jetson shall continue polling LC status using `0x11`.

The Jetson shall not start robot operation while the lift is not available.

Since serial communication polling is continuous / real-time, the Jetson does not need to wait 1 minute before checking again. The Jetson may retry the `0x12` Start Automatic Service Preparation command until the LC acknowledges it or the expected `0x11` state transition is observed. After that, the Jetson continues polling `0x11` and will detect `Ready for Jetson/Robot Control` as soon as the LC reports that the lift is ready.

Maximum Servicing Duration

The automatic maintenance workflow shall have a maximum servicing duration to prevent the lift from being held for maintenance for too long.

The initial proposed maximum duration for the full servicing process is `1 hour`.

This maximum duration is measured from the time the Jetson sends the `0x12` Start Automatic Service Preparation request until the LC reports:

```
Automatic Service Session State = Service Complete / Normal Operation
Restored
```

The `1 hour` maximum duration is an initial proposed value and shall be further discussed and confirmed.

The purpose of this limit is to ensure that the automatic servicing workflow does not occupy the lift for an excessive period, especially if the system is deployed in residential buildings where residents may need to use the lift during late-night hours.

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2. LC Preparation Monitoring

While the LC is performing automatic service preparation, the Jetson shall continue holding Automatic Service Request = ON through periodic `0x12` hold frames and continuously poll `0x11` for LC status.

The Jetson shall continue waiting if the LC reports any preparation-related state such as:

```
Preparation Request Received
Checking Readiness
Entering Robot Maintenance Mode
Moving to Top Service Floor
```

The Jetson shall proceed only when:

```
Automatic Service Session State = Ready for Jetson/Robot Control
```

If the LC reports a failed or aborted state:

```
Automatic Service Session State = Failed
or
Automatic Service Session State = Aborted
```

Then the Jetson shall stop the automatic maintenance workflow and refer to the `Automatic Service Abnormal Reason` reported by the LC.

If Lift Abnormal Mode is detected at any point, the Jetson shall stop the current maintenance workflow immediately.

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3. Start Robot Inspection Workflow

Once the LC reports `Ready for Jetson/Robot Control`, the Jetson may begin the robot inspection workflow.

Before starting robot movement, the Jetson shall confirm:

```
Automatic Service Session State = Ready for Jetson/Robot Control
Jetson Operation Permitted = 1 (ON)
Robot Maintenance Mode = 1 (ON)
Lift Abnormal Mode = 0 (OFF)
Robot is ready
```

During the robot inspection workflow, the Jetson shall continue polling `0x11`.

If `Automatic Service Session State` is no longer `Ready for Jetson/Robot Control`, `Jetson Operation Permitted` turns OFF, `Robot Maintenance Mode` turns OFF, or `Lift`

`Abnormal Mode` turns ON, the Jetson shall immediately stop the current robot operation and stop the inspection workflow.

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4. Move Below Top Floor to Inspection Position

After the LC has completed automatic service preparation and the lift is at the top service floor, the Jetson begins positioning the car for the first landing door inspection.

The first inspection point is not exactly at the top floor. The car moves downward from the top floor until the selected positioning reference reaches the required stopping condition.

The Jetson shall use the current movement direction, current floor, service floor sequence, and selected positioning reference to determine the current inspection point.

The Jetson may use `0x10` distance data as a coarse positioning reference or sanity check during manual down positioning.

Positioning Reference Options

Primary method:

Use `0x10` Distance from Car Top to Current Floor Sill.

Fallback method:

If the `0x10` Distance from Car Top to Current Floor Sill is unreliable, unavailable, or not suitable for final positioning, use the current floor information from `0x11` and camera-based detection of a defined inspection point for each service floor.

```
Function Code: 0x10
Read: Distance from Car Top to Current Floor Sill
```

The Jetson sends `0x12` with Manual Down held ON.

```
Function Code: 0x12
Destination Floor Byte = 0x00
Manual Down = 1 (ON)
Manual Up = 0 (OFF)
Reserved Door Command Bits = 0 (OFF)
Hall Call Disable = 1 (ON)
Automatic Service Request = 1 (ON)
Automatic Service Request Type = 0x01
Request Type Meaning = Hold Robot Maintenance Operation
```

During manual down positioning, the Jetson shall continue polling `0x11`.

The Jetson shall continue manual down movement only while:

```
Automatic Service Session State = Ready for Jetson/Robot Control
Jetson Operation Permitted = 1 (ON)
Robot Maintenance Mode = 1 (ON)
Lift Abnormal Mode = 0 (OFF)
```

Manual Down remains `ON` while the selected positioning reference has not reached the required stopping condition.

If the distance-based method is used, the Jetson keeps `Manual Down = ON` until:

```
abs(current_car_top_to_floor_sill_distance -
target_car_top_to_floor_sill_distance) <= configured_tolerance
```

If the camera-based method is used, the Jetson keeps `Manual Down = ON` until the camera detects the defined inspection point for the current service floor with sufficient confidence.

Once the stopping condition is reached, the Jetson sends another `0x12` command with Manual Down turned OFF.

```
Function Code: 0x12
Destination Floor Byte = 0x00
Manual Down = 0 (OFF)
Manual Up = 0 (OFF)
Reserved Door Command Bits = 0 (OFF)
Hall Call Disable = 1 (ON)
Automatic Service Request = 1 (ON)
Automatic Service Request Type = 0x01
Request Type Meaning = Hold Robot Maintenance Operation
```

The lift is now stationary at the inspection height.

If the selected positioning reference cannot be confirmed within the expected movement range, configured timeout, or LC-side safety limit, the Jetson shall turn Manual Down OFF, stop the inspection workflow, and wait for recovery or manual intervention.

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5. Robot Inspection Hold State

At the inspection position, the robot performs the required landing door inspection or servicing task.

During this period, the Lift Controller should not execute independent lift movement, door movement, or new movement-related commands.

Allowed during this state:

```
0x11 status polling
0x10 distance reading for logging
```

Not allowed during this state:

```
Manual Up
Manual Down
Destination floor command
Reserved Door Command Bits = 1
Any lift movement
Any door movement
```

The Jetson keeps the lift in the following command state:

```
Hall Call Disable = 1 (ON)
Manual Down = 0 (OFF)
Manual Up = 0 (OFF)
Reserved Door Command Bits = 0 (OFF)
Automatic Service Request = 1 (ON)
Automatic Service Request Type = 0x01
Request Type Meaning = Hold Robot Maintenance Operation
```

During robot inspection, the LC session state is expected to remain:

```
Automatic Service Session State = Ready for Jetson/Robot Control
Jetson Operation Permitted = 1 (ON)
Robot Maintenance Mode = 1 (ON)
Lift Abnormal Mode = 0 (OFF)
```

If `Automatic Service Session State` is no longer `Ready for Jetson/Robot Control`, `Jetson Operation Permitted` turns OFF, `Robot Maintenance Mode` turns OFF, or `Lift Abnormal Mode` turns ON, the Jetson shall immediately stop the current robot operation and stop the inspection workflow.

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6. Repeat for Lower Service Floors

After the robot completes the inspection at the current landing door, the Jetson prepares to move to the next lower service floor.

The Jetson checks the service floor configuration and identifies the next valid service floor below the current inspection level.

The Jetson shall only stop when the selected positioning reference corresponds to the next valid service floor. If the selected positioning reference does not correspond to a valid service floor inspection point, the Jetson shall continue moving downward or stop according to the configured safety limit.

For each valid service floor, the Jetson repeats:

```
1. Poll LC status using 0x11.
2. Confirm Automatic Service Session State = Ready for Jetson/Robot Control.
3. Confirm Jetson Operation Permitted is ON, Robot Maintenance Mode is ON,
and Lift Abnormal Mode is OFF.
4. Send 0x12 with Manual Down = 1 (ON).
5. Monitor the selected positioning reference using `0x10` Distance from Car
Top to Current Floor Sill and/or camera-based inspection point detection.
6. When the selected stopping condition is reached, send Manual Down = 0
(OFF).
7. Hold lift stationary.
8. Allow robot inspection.
9. Continue polling 0x11 during inspection.
```

The sequence continues from the top service floor downward until the lowest required service point is completed, normally the 2nd floor.

The system does not attempt to service below the 2nd floor because the lift car cannot move below the 1st floor.

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7. Complete Automatic Service Session

After the final inspection point is completed, the Jetson clears `Automatic Service Request` to OFF. The drop from `Automatic Service Request = ON` to `Automatic Service Request = OFF` is the completion request for the current robot maintenance operation.

The final inspection point may be at any service floor depending on the inspection sequence.

```
Function Code: 0x12
Destination Floor Byte = 0x00
Manual Down = 0 (OFF)
Manual Up = 0 (OFF)
Reserved Door Command Bits = 0 (OFF)
Hall Call Disable = 0 (OFF)
Automatic Service Request = 0 (OFF)
Automatic Service Request Type = 0x00
Request Type Meaning = End Robot Maintenance Request / Return to Normal
```

The Jetson shall resend this request-cleared command until either:

1. the LC acknowledges the command, or
2. the Jetson observes the expected completion-related state transition through 0x11 status polling.

After the request-cleared command is acknowledged or the expected state transition is observed, the Jetson shall stop retrying the request change and continue monitoring completion using 0x11.

After sending this command, the Jetson shall only poll `0x11`.

The LC shall handle the automatic service completion sequence internally.

```
- Restore Auto mode from the current lift position
- Level the lift as required
- Re-enable hall calls / normal passenger operation
- Report service complete
```

The Jetson shall not send Manual Up, Manual Down, destination floor, or door commands while the LC automatic service completion sequence is in progress.

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8. Completion Monitoring

After the Jetson clears Automatic Service Request to OFF, the Jetson shall continuously poll 0x11 while the LC performs automatic service completion.

The Jetson shall continue waiting if the LC reports any completion-related state such as:

```
Completion Request Received
Returning to Auto Mode
```

The automatic service session is complete when the LC reports:

```
Automatic Service Session State = Service Complete / Normal Operation
Restored
Robot Maintenance Mode = 0 (OFF)
Lift Abnormal Mode = 0 (OFF)
```

The lift is now returned to normal passenger operation.

Error / Abnormal Handling

LC No Response

If the LC does not reply within `100 ms`, the Jetson retries communication.

```
General timeout:
- Retry communication.
- Raise alert if communication is not restored within 5 minutes.
```

If the timeout occurs during Jetson-controlled manual down movement:

```
Timeout during manual movement:
- Immediately attempt to send Manual Up = OFF and Manual Down = OFF.
- Continue retrying communication.
- Raise alert if communication is not restored within 5 minutes.
```

Lift Not Available at 3am

If the LC reports that the lift is not available for automatic service:

```
Automatic Service Session State = Not Available
```

Then the Jetson shall:

```
1. Continue polling LC status using `0x11`.
2. Do not start robot operation while the lift is not available.
3. Proceed only when the LC reports `Ready for Jetson/Robot Control`.
4. Stop the workflow if the LC reports `Failed` or `Aborted`.
```

Maximum Servicing Duration Exceeded

If the automatic maintenance workflow exceeds the configured maximum servicing duration, the Jetson shall stop the current maintenance workflow and request the LC to return the lift to normal passenger operation.

Expected Jetson behaviour:

```
1. Stop the current robot operation if one is active.
2. Clear Manual Up and Manual Down commands if manual movement is active.
3. Stop further robot inspection for the remaining service floors.
4. Record the automatic service session result in the Jetson logs as
incomplete due to maximum servicing duration exceeded.
5. Clear Automatic Service Request to OFF to request LC return-to-normal
handling.
```

```
6. Continue polling LC status using `0x11`.
7. Wait until the LC reports `Service Complete / Normal Operation Restored`.
```

The intended recovery action is for the LC to return the lift to normal passenger operation after the maximum servicing duration is exceeded.

The initial proposed maximum servicing duration is `1 hour`, measured from the time the Jetson sends the `0x12` Start Automatic Service Preparation request until the LC reports:

```
Automatic Service Session State = Service Complete / Normal Operation
Restored
```

The `1 hour` maximum duration is an initial proposed value and shall be further discussed and confirmed.

If the LC cannot safely return the lift to normal operation due to an active safety fault, interlock condition, Lift Abnormal Mode, or other LC-side abnormal condition, the LC safety logic shall take priority. In this case, the Jetson shall continue polling `0x11`, keep the robot stopped, and wait for recovery or manual intervention.

LC Preparation or Completion Failed

If the LC reports:

```
Automatic Service Session State = Failed
or
Automatic Service Session State = Aborted
```

The Jetson shall read the `Automatic Service Abnormal Reason` from `0x11`.

The Jetson shall stop the current automatic maintenance workflow and wait for recovery or manual intervention.

Possible abnormal reasons include:

```
Lift Not Available
Door Not Closed
Passenger / Service Conflict
Lift Abnormal Mode Detected
Failed to Enter Robot Maintenance Mode
Failed to Return to Auto Mode
Timeout
Safety / Interlock Condition
Internal LC Error
```

Technician Manual Intervention During Inspection

This scenario applies when a technician manually intervenes while the Jetson is already running the robot inspection workflow.

The exact LC-side trigger for technician/manual takeover is to be confirmed with the engineering team.

Expected LC status update:

```
Automatic Service Session State = Aborted
Automatic Service Abnormal Reason = Manual Technician Intervention / Manual
Service Takeover
```

When technician/manual takeover is detected, the LC shall ignore or override further Jetson lift movement commands according to LC safety logic.

Expected Jetson behaviour:

```
1. Jetson detects the LC status change through 0x11.
2. Jetson immediately stops the current robot operation.
3. Jetson stops the current inspection workflow.
4. Jetson stops sending lift movement, manual down, or robot movement
commands.
5. System waits for technician/manual recovery.
6. Automatic servicing shall not resume automatically.
```

Robot Fault During Inspection

This scenario applies when the robot encounters a fault while the Jetson is already running the inspection workflow.

Possible triggers include:

```
- Robot collision detected
- Robot brake clamp
- Robot motion fault
- Robot unable to continue current task
```

Expected Jetson behaviour:

```
1. Jetson detects the robot fault from the robot controller.
2. Jetson immediately stops the current robot operation.
```

```
3. Jetson stops the current inspection workflow.
4. Jetson immediately clears active lift movement commands if manual
movement is active.
5. Jetson sends 0x12 Abort Automatic Service request to the LC.
```

Abort request:

```
Function Code: 0x12
Destination Floor Byte = 0x00
Manual Up = 0 (OFF)
Manual Down = 0 (OFF)
Reserved Door Command Bits = 0 (OFF)
Hall Call Disable = 1 (ON)
Automatic Service Request = 1 (ON)
Automatic Service Request Type = 0x02
Request Type Meaning = Abort Robot Maintenance Operation
```

Expected LC status update:

```
Automatic Service Session State = Aborted
Automatic Service Abnormal Reason = Abort Requested by Jetson
```

The detailed robot fault reason shall be recorded in the Jetson logs.

The LC shall hold the lift in a safe stopped / Robot Maintenance state according to LC safety logic.

The Jetson shall continue sending the abort request until the LC acknowledges the command or `0x11` reports `Automatic Service Session State = Aborted`.

Operator or technician shall attend site to manually recover the robot. Automatic servicing shall not resume until the robot fault is cleared and the system is manually confirmed safe to restart.

Lift Abnormal Mode Detected

If Lift Abnormal Mode is detected at any point during automatic service:

```
Lift Abnormal Mode = 1 (ON)
```

The Jetson shall:

```
1. Immediately stop the current robot operation if one is active.
2. Immediately attempt to clear Manual Up and Manual Down commands if manual
```

```
movement is active.
3. Stop sending further lift movement commands.
4. Stop the automatic maintenance workflow.
5. Keep polling LC status.
6. Wait for recovery or manual intervention.
```

LC State Not Ready During Inspection

During Jetson-controlled manual down positioning or robot inspection, the expected LC state is:

```
Automatic Service Session State = Ready for Jetson/Robot Control
Jetson Operation Permitted = 1 (ON)
Robot Maintenance Mode = 1 (ON)
Lift Abnormal Mode = 0 (OFF)
```

If any of these conditions are no longer true during inspection, the Jetson shall treat it as an abnormal condition.

Expected Jetson behaviour:

```
1. Immediately stop the current robot operation.
2. Immediately stop the inspection workflow.
3. If manual down movement is active, send Manual Down = 0 (OFF).
4. Stop sending further lift movement or robot movement commands.
5. Read the Automatic Service Abnormal Reason from 0x11 if available.
6. Wait for recovery or manual intervention.
```

Jetson Communication Lost

This scenario applies if the LC does not receive expected Jetson communication while an automatic service session is active.

The exact timeout duration and LC-side handling shall be discussed with the engineering team.

Possible LC behaviour to be confirmed:

```
- Detect Jetson communication loss after a configured timeout.
- Stop or hold the automatic service session according to LC safety logic.
- Report Automatic Service Session State = Aborted or Failed.
- Report Automatic Service Abnormal Reason = Jetson Communication Lost.
```

Manual Down Safety

During manual down movement:

```
Manual Down = ON only while moving toward the selected stopping condition
for the current inspection point.
Manual Down = OFF immediately when the selected stopping condition is
reached.
```

The LC should enforce safety limits to prevent continued lift movement if the Jetson fails to clear the Manual Down bit.

Manual Down shall not remain ON if Lift Abnormal Mode is detected, Jetson Operation Permitted turns OFF, or Robot Maintenance Mode turns OFF.

Possible LC-side safety limits to discuss with engineering:

```
- Maximum manual down command duration
- Maximum movement distance during manual down
- Stop condition if distance reading is invalid or not changing
- Stop condition if communication with Jetson is lost
```